LLM EVALUATION

DATASETS

METRICS

EVALUATION

ANALYSIS

EXPERIMENTS

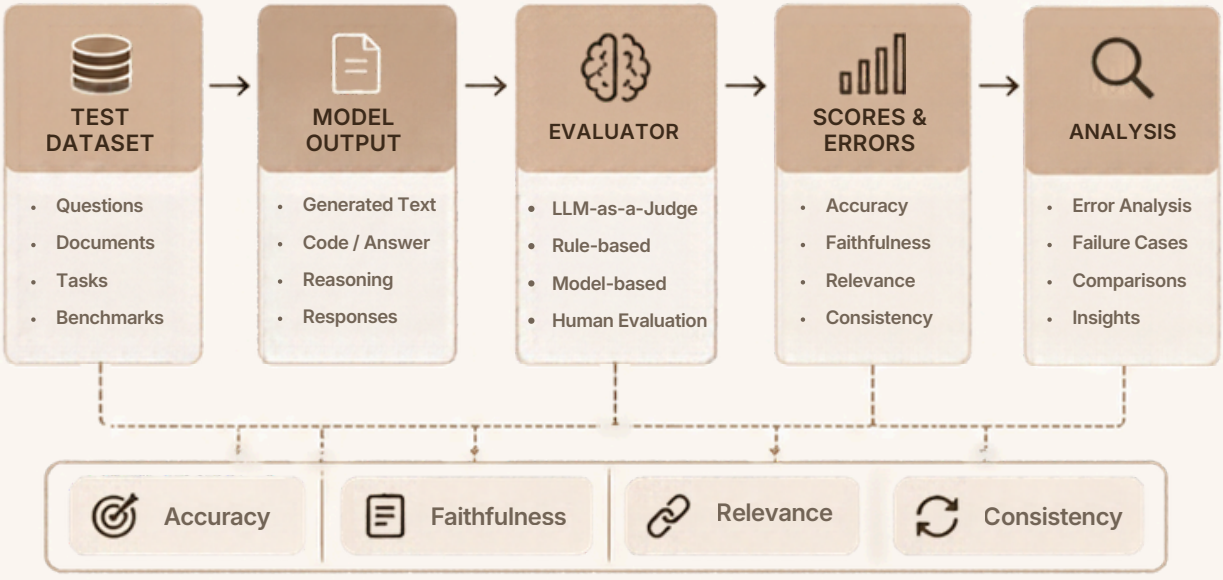
MEASURE
UNDERSTAND
IMPROVE

LLM EVALUATION · CHAPTER 02

WHAT ARE LLM EVALUATIONS?

Measuring Quality, Reliability, and Model Performance

A practical guide to evaluation datasets, metrics, LLM-as-a-Judge, failure analysis, and repeatable experiences.



WHAT THIS SERIES COVERS

- Evaluation Design
- Metrics & Rubrics
- LLM-as-a-Judge
- Regression & Error Analysis

PREMIUM REVISION NOTES

CONCEPTS · IMPLEMENTATION · DIAGRAMS · INTERVIEW PREP

What Are LLM Evaluations?

RECAP — CHAPTER 1

Chapter 1 covered the motivation: why LLM evaluations are needed, why they matter, and how they differ from classical software testing. This chapter moves from *why* to *what*: it defines LLM evals precisely and splits them into two types, model evals and application evals.

1. What Are LLM Evals?

Definition

LLM evals are systematic, repeatable tests used to judge an LLM, or an LLM-powered system, against a clear criteria. They are applied to bare LLMs and also to LLM-based applications such as chatbots. Three characteristics define them: they are *systematic*, *repeatable*, and tied to *clear criteria*.

Property	Meaning	Campus X chatbot example
Systematic	No “vibe testing” (asking five questions and feeling it looks fine). You build proper datasets and try to cover every edge case.	Sample 100 real user chats with the system and build a dataset from them, so tests reflect real-world behaviour.
Repeatable	After changing the prompt, model, retriever, or chunking strategy, the same dataset can be applied to the new version in exactly the same way.	Run the same test dataset on version 1 and version 2 and compare scores to see whether the system improved.
Clear criteria	Evaluation depends on *what* you judge against. Without criteria it is vibe testing; with criteria it is proper evaluation.	Answer is correct; explanation is simple; answer comes from our own course content; response is safe (no unsafe content, abuse, or threatening tone).

IMPORTANT

Repeatability is what makes comparison possible. Because the test dataset stays fixed, differences in performance between versions can be attributed to the system change.

↻ Quick Revision — Definition of LLM Evals

- LLM evals are systematic, repeatable tests that judge an LLM or LLM-powered system against clear criteria.
- Systematic means proper datasets covering edge cases, not ad-hoc vibe testing.
- Repeatable means the same dataset can be applied to any version of the system to compare versions.
- Criteria (correctness, groundedness, simplicity, safety) define what "good" means.

2. LLM Evals Are More Than Metrics

The Common Confusion

Coming from classical machine learning and deep learning, "evaluation" usually means metrics such as accuracy, precision, and recall. It is tempting to assume that LLM evals are simply a set of metrics. That is not correct.

What an Eval Actually Includes

An LLM eval is the complete testing setup built to test an LLM or LLM-based system. For a RAG chatbot, as shown in Figure 2.1, the setup covers all of the following:

- **What is being tested** — for example, the retriever inside a RAG chatbot.
- **The criteria** — for example, retriever accuracy.
- **The test dataset** prepared for the evaluation.
- **When the test runs** — offline, or after deployment to production.
- **The tools used** — for example, Ragas for a RAG application.

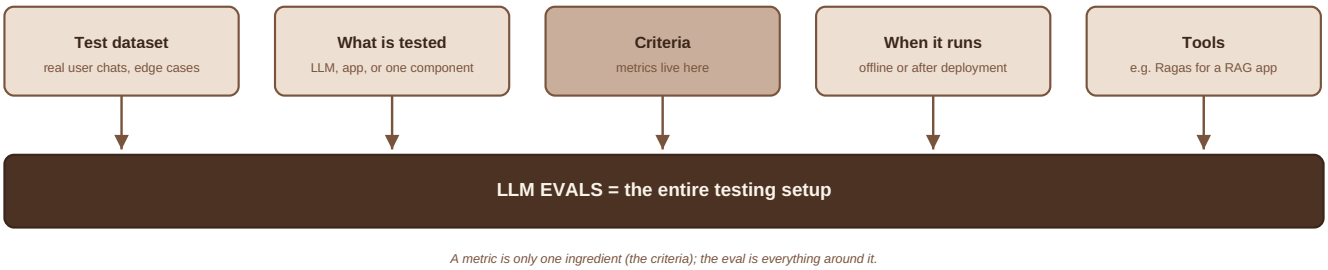


Figure 2.1 — An LLM eval is the whole testing setup; metrics only supply the criteria.

COMMON MISTAKE

Treating "LLM evals" as a synonym for "metrics." If asked what LLM evals are, answer that they are not just metrics but the entire testing setup: what is tested, against what criteria, when, and with which tools.

↻ Quick Revision — Evals vs Metrics

- LLM evals are not only metrics; they are the entire testing setup.
- The setup includes the subject under test, the criteria, the dataset, the timing (offline or in production), and the tools.
- A metric such as retriever accuracy is one part of an eval, not the eval itself.

3. What Evals Help You Decide

The goal of an LLM eval is **not to produce a score**. Its goal is to answer practical questions about the system:

- Can this model be used for a particular task or application?
- Is this system good enough to ship, i.e. can it go into production?
- Did prompt version 2 improve over prompt version 1?
- Is the RAG answer grounded in the retrieved context?
- Is the agent completing its task correctly?
- Is the chatbot safe for real users?
- Is latency under control?

> **REAL WORLD INSIGHT**
>
> Each question above maps to a decision an engineer must actually make. A score is useful only when it helps answer one of them.

↻ Quick Revision — Purpose of Evals

- Evals exist to answer practical questions, not just to output a number.
- Typical questions: is it good enough to ship, did the new prompt improve, is the answer grounded, is it safe, is latency acceptable.

4. Two Types of LLM Evals

Model Evals vs Application Evals

Following the definition, which mentions both an LLM and an LLM-powered system, evals can be divided into two parts.

	Model evals	Application evals
Evaluates	The LLM itself	The entire LLM-based application, or a specific component inside it
Core question	“Can the model do this?”	“Will our product work correctly?”
Typical moment	When a new LLM is released	While building and running your own application
Method	Benchmarks and leaderboards	Datasets, criteria, and tooling designed for your product
Who mainly does it	Large frontier labs	AI engineers building applications

IMPORTANT

“Model evals” and “application evals” are **not official industry terms**. They were introduced in the lecture to simplify the topic, so do not cite them as standard terminology. The industry calls both simply “LLM evals”, and people infer from the use case which one is meant.

↻ Quick Revision — Two Types

- Model evals test the LLM; application evals test the whole LLM-based application or one of its components.
- The two terms are teaching labels from the lecture, not official industry terms.
- In industry, both are usually just called LLM evals.

5. Model Evals

Definition

Model evals evaluate the model itself. The main idea is to test and evaluate a model’s **capabilities**. When a new LLM is released, its capabilities are tested, benchmarked, and documented, which is why launches cite a particular benchmark or leaderboard where the model came out on top with a certain accuracy.

Eight Capabilities Commonly Tested

Current LLMs are mainly tested on eight capability categories, each measured through benchmarks. The benchmarks below are examples named in the lecture; the popular ones are covered in the next lecture.

Capability	What is tested	Example benchmark
Reasoning	Can the model solve a problem by thinking step by step?	MMLU (general knowledge and reasoning)
Knowledge	Basic world and general knowledge up to the training cutoff date	MMLU (questions on science, history, law, medicine, and more)
Basic maths	Can it solve maths problems?	GSM8K (grade-school maths)
Coding	Can it write code?	SWE-bench, HumanEval
Instruction following	Given e.g. ten instructions, does it follow all of them one after another?	IFEval
Long-context handling	Can it extract correct answers from a very large context?	Needle in a Haystack
Multimodal understanding	Can it understand, and output, images, text, and sound?	MMMU
Tool use	Can it use tools properly?	No benchmark named in the lecture

How AI Engineers Use Model Evals

Evaluating, benchmarking, and documenting a newly released LLM is the job of the large frontier labs, not of an application engineer. An AI engineer is unlikely to run model evals in practice, but still needs to know:

1. what model evaluation is,
2. what benchmarks are and which are the top ones,
3. how to read benchmark results.

This knowledge supports **model selection** at the start of a project: whether to use an OpenAI model, an Anthropic model, or an open-source LLM. That decision comes out of model evals.

> **INTERVIEW TIP**
>
> Be ready to explain what a benchmark is and to name a benchmark for each capability (for example GSM8K for maths and SWE-bench for coding), and to say how benchmark results guide model choice for an application.

↻ Quick Revision — Model Evals

- Model evals test an LLM's capabilities using benchmarks and leaderboards.
- Eight capabilities: reasoning, knowledge, basic maths, coding, instruction following, long-context handling, multimodal understanding, tool use.
- Frontier labs run these evals when releasing a model; application engineers mostly read and use the results.
- Benchmark results support the decision of which LLM to use in a project.

6. Application Evals

Application evals are the category you should study carefully and will do in practice, since an AI engineer's job is to build LLM-based applications and therefore also to evaluate them. They are the main focus of this course.

Why They Exist: The LLM Is One Component

Beginners often assume the LLM is everything in an LLM application. In a real application it is just one component, the "brain". Around it sit many other parts, shown in Figure 2.2, that must also work correctly: the user interface, system prompt, tools and APIs, orchestration code (for example branching and parallel control flow in LangGraph), guardrails, output parsers, memory and context, and, in a RAG system, retrieval, an embedding model, and vector databases. After deployment there is also monitoring and a feedback loop.

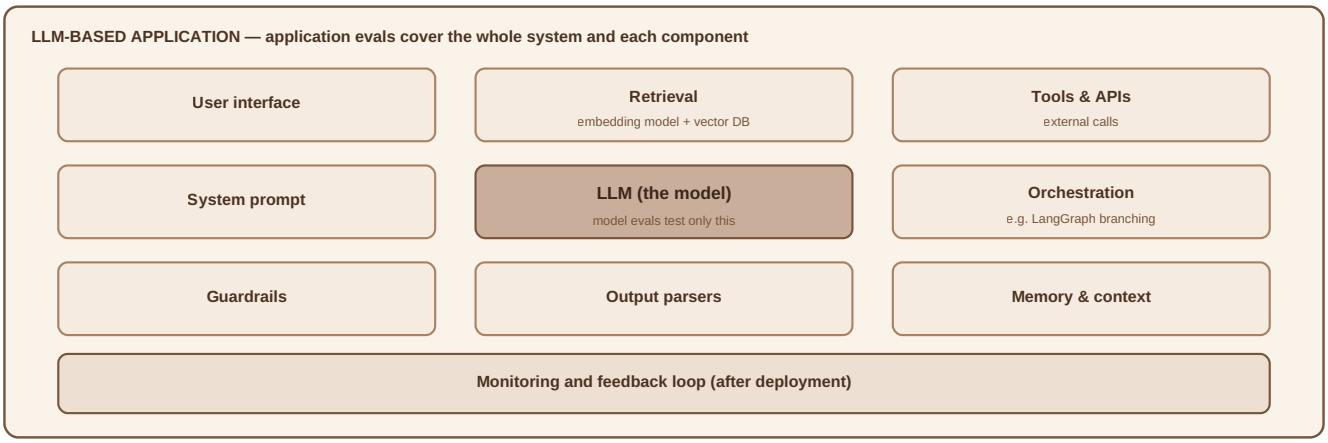


Figure 2.2 — The LLM is one component of an LLM-based application; application evals cover the rest too.

Smartphone Analogy

A chip maker such as Snapdragon or MediaTek publishes benchmark scores to show how strong its processor is. That does not guarantee a good smartphone: the camera system, operating system, sound system, graphics, battery, and screen all matter, and each is tested too. Likewise, frontier labs evaluate the LLM for us (the processor), but evaluating the system we build around it is our responsibility as AI engineers.

What Application Evals Assess

Application evals assess the behaviour and performance of an LLM-powered application, either at the level of the **entire system** or of a **specific component** inside it. They do not ask "can the model do this?" (model evals); they ask "will our product work correctly?"

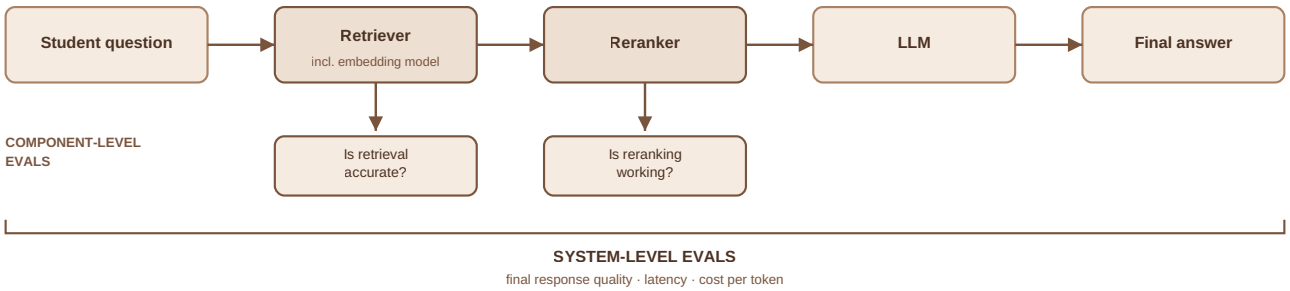


Figure 2.3 — Illustrative RAG chatbot: system-level evals judge the whole; component-level evals judge parts.

Level	Example for a RAG chatbot
System level	Quality of the final response, latency, cost per token
Component level	Is the retriever working? Is the embedding model working? Is the reranker working?

Example: Campus X Chatbot

For a chatbot built for Campus X, application evals help answer:

- Was the student's question answered correctly?
- Was the course material used properly?
- Was the answer faithful?
- Was the answer easy enough for a beginner?
- Did hallucination occur?
- Was the answer quick?
- Is the chatbot safe?

IMPORTANT

Model evals tell you how capable the model is; they do not tell you whether the application built around it works. Both the whole system and each component need their own evaluation.

INTERVIEW TIP

Per the lecture, when a video or article says "LLM evaluation" you can assume, most likely about 99% of the time, that it means application evaluation. The rest of this course follows that convention.

↻ Quick Revision — Application Evals

- The LLM is one component of an application, so model evals alone are not enough.
- Application evals assess behaviour and performance at system level and at component level.
- System level: final response, latency, cost per token. Component level: retriever, embedding model, reranker.
- They answer product questions: correctness, use of course material, faithfulness, beginner-friendliness, hallucination, speed, safety.
- Smartphone analogy: a strong processor does not guarantee a good phone; frontier labs test the LLM, engineers test the system.

7. Where the Course Goes Next

So far the lecture covered **why** we study evals (previous chapter) and **what** LLM evals are, together with their two types. The next step is **how** LLM evaluations are actually done. That “how” will be taught from the **application eval** perspective, not the model eval perspective. Model evals get one dedicated lecture, covering the most popular benchmarks and how they are applied to LLMs and read, so that you have a working understanding of them.

↻ Quick Revision — Course Flow

- Why and what are now covered; how is next.
- The how is taught from the application eval perspective.
- Model evals get a single dedicated lecture on popular benchmarks.

K. Key Takeaways

- **Definition:** LLM evals are systematic, repeatable tests that judge an LLM or LLM-powered system against clear criteria.
- **Evals vs metrics:** an eval is the entire testing setup — subject, criteria, dataset, timing, and tools — not just metrics.
- **Purpose:** evals answer practical questions such as whether the system is good enough to ship or whether a new prompt improved it, not just produce scores.
- **Two types:** model evals test the LLM; application evals test the LLM-based application or its components. These are teaching labels, not official terms.
- **Model evals:** test eight capabilities through benchmarks, mainly run by frontier labs; engineers use them to read results and choose a model.
- **Application evals:** the focus of this course, since the LLM is only one component of an application; they work at system and component level.
- **Convention:** "LLM evaluation" usually means application evaluation.

R. Revision Sheet

LLM Evals

Systematic, repeatable tests that judge an LLM or an LLM-powered system against clear criteria.

Systematic / Repeatable / Criteria

Proper datasets covering edge cases; same dataset reusable on any version; defined criteria such as correctness, groundedness, simplicity, and safety.

Evals vs Metrics

A metric is one part. The eval is the complete setup: what is tested, criteria, dataset, when it runs, and tools.

Model Evals

Evaluate the LLM's capabilities using benchmarks and leaderboards; mainly done by frontier labs.

Eight Capabilities

Reasoning, knowledge, basic maths, coding, instruction following, long-context handling, multimodal understanding, tool use.

Application Evals

Evaluate an LLM-based application at system level or component level; they ask whether the product will work.

System vs Component Level

System: final response, latency, cost per token. Component: retriever, embedding model, reranker.

I. Interview Questions

Q1. Define LLM evals and explain the three characteristics the lecture highlights.

Q2. Why is saying “LLM evals are just metrics” incorrect? What does a complete eval setup include?

Q3. What does repeatability enable when you change the prompt, model, retriever, or chunking strategy?

Q4. How would you build a systematic test dataset for a chatbot you are developing?

Q5. Distinguish model evals from application evals. Are these official industry terms?

Q6. Name the eight capabilities commonly tested in model evals and a benchmark for several of them.

Q7. Why is an AI engineer unlikely to run model evals, and why must they still understand benchmarks?

Q8. Why is the LLM only one component of an LLM application, and why does that make application evals necessary?

Q9. Explain the smartphone-processor analogy in the context of model vs application evals.

Q10. For a RAG chatbot, what would you evaluate at the system level versus the component level?

F. Further Reading

- Benchmark papers for MMLU, GSM8K, HumanEval, SWE-bench, IFEval, and MMMU (search by benchmark name; the next lecture covers the popular ones).
- Ragas documentation — the tool the lecture names for evaluating RAG applications.